

Examples of how to build top-down and bottom-up parsers given a grammar

Note that this grammar is ambiguous. While this won't affect how we will build our parsers here, we always want to first make our grammar non-ambiguous

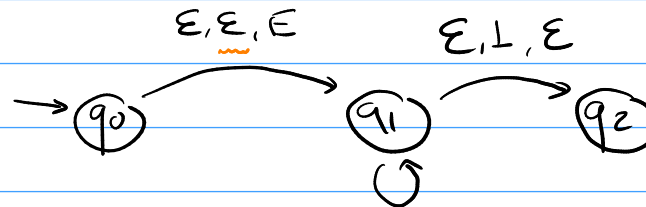
(1) grammar $E ::= \underline{E+E} \mid \underline{E \times E} \mid \underline{(E)} \mid \underline{num}$

① ② ③ ④

(2) top-down parser for this grammar:

initial non terminal

initial element in our stack



The pda is in black and my annotations are in colour

4 production rules in our grammar so 4 transitions

1 transition for each terminal symbol

{	$E, E, E+E$	①
	$E, E, E \times E$	②
	$E, E, (E)$	③
	E, E, num	④
	$+, +, E$	⑤
	$\times, \times, E$	⑥
	$(, (, E$	⑦
	$),), E$	⑧
	num, num, E	⑨

We're using the notation from pda's - pda

(see the underlined E)

Can we parse $0+1$?

$(q_0, 0+1, \perp) \rightarrow (q_1, 0+1, \underline{E}\perp)$ transition ①
 $\rightarrow (q_1, 0+1, \underline{E+E}\perp)$ ④
 $\rightarrow (q_1, 0+1, \underline{0+E}\perp)$ ⑨
 $\rightarrow (q_1, +1, \underline{+E}\perp)$ ⑤
 $\rightarrow (q_1, 1, \underline{E}\perp)$ ④
 $\rightarrow (q_1, 1, \underline{1}\perp)$ ⑧
 $\rightarrow (q_1, \epsilon, \perp)$
 $\rightarrow (q_2, \epsilon, \epsilon)$

Note that we only ever pop the top of the stack!

— yes we can parse this expression

Can we parse $0+$?

$(q_0, 0+, \perp) \rightarrow (q_1, 0+, \underline{E}\perp)$ ①
 $\rightarrow (q_1, 0+, \underline{E+E}\perp)$ ④
 $\rightarrow (q_1, 0+, \underline{0+E}\perp)$ ⑨
 $\rightarrow (q_1, +, \underline{+E}\perp)$ ⑤
 $\rightarrow (q_1, \epsilon, \underline{E}\perp)$

— stuck — we cannot parse this expression

Can we parse $0+1 \times 2$?

alternatively we could do this

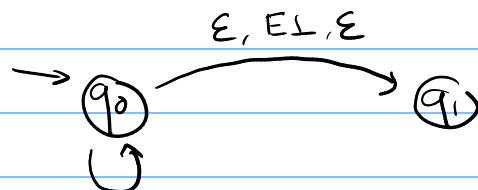
$(q_0, 0+1 \times 2, \perp) \rightarrow (q_1, 0+1 \times 2, E\perp)$	$\rightarrow (q_1, 0+1 \times 2, E\perp)$
$\rightarrow (q_1, 0+1 \times 2, E+E\perp)$	$\rightarrow (q_1, 0+1 \times 2, E \times E\perp)$
$\rightarrow (q_1, 0+1 \times 2, 0+E\perp)$	$\rightarrow (q_1, 0+1 \times 2, E+E \times E\perp)$
$\rightarrow (q_1, +1 \times 2, +E\perp)$	$\rightarrow (q_1, 0+1 \times 2, 0+E \times E\perp)$
$\rightarrow (q_1, 1 \times 2, E\perp)$	$\rightarrow (q_1, +1 \times 2, +E \times E\perp)$
$\rightarrow (q_1, 1 \times 2, E \times E\perp)$	$\rightarrow (q_1, 1 \times 2, E \times E\perp)$
$\rightarrow (q_1, 1 \times 2, 1 \times E\perp)$	$\rightarrow (q_1, 1 \times 2, 1 \times E\perp)$
$\rightarrow (q_1, \times 2, \times E\perp)$	$\rightarrow (q_1, \times 2, \times E\perp)$
$\rightarrow (q_1, 2, E\perp)$	$\rightarrow (q_1, 2, E\perp)$
$\rightarrow (q_1, 2, 2\perp)$	$\rightarrow (q_1, 2, 2\perp)$
$\rightarrow (q_1, \epsilon, \perp)$	$\rightarrow (q_1, \epsilon, \perp)$
$\rightarrow (q_2, \epsilon, \epsilon)$	$\rightarrow (q_2, \epsilon, \epsilon)$

we've parsed $0+[1 \times 2]$

we parsed $[0+1] \times 2$

This illustrates the fact that the grammar is ambiguous

(3) bottom-up parser for this grammar:



reduces, 1 for each production rule

- $E, E+E, E$ ①
- $E, E \times E, E$ ②
- $E,)E(, E$ ③
- E, num, E ④

we're again using the notation from ptas.pdf

shifts, 1 for each terminal

- $+, E, +$ ⑤
- $\times, E, \times$ ⑥
- $(, E, ($ ⑦
- $), E,)$ ⑧
- $\text{num}, E, \text{num}$ ⑨

see how we have $)E($ and $\text{num}(E)$ because reading (0) for example, we would
 shift (onto the stack: $(\perp$
 shift 0 onto the stack: $0(\perp$
 reduce 0 to E: $E\perp$
 shift) onto the stack: $)E\perp$

Can we parse $0+1$

$(q_0, 0+1, \perp) \xrightarrow{\text{shift}} \textcircled{3}$
 $\rightarrow (q_0, +1, 0\perp) \xrightarrow{\text{reduce}} \textcircled{4}$
 $\rightarrow (q_0, +1, E\perp) \xrightarrow{\text{shift}} \textcircled{5}$
 $\rightarrow (q_0, 1, +E\perp) \xrightarrow{\text{shift}} \textcircled{9}$
 $\rightarrow (q_0, \varepsilon, 1+E\perp) \xrightarrow{\text{reduce}} \textcircled{4}$
 $\rightarrow (q_0, \varepsilon, E+E\perp) \xrightarrow{\text{reduce}} \textcircled{1}$
 $\rightarrow (q_1, \varepsilon, \varepsilon)$
accepted

Can we parse $0+$?

$(q_0, 0+, \perp) \xrightarrow{\text{shift}} \textcircled{3}$
 $\rightarrow (q_0, +, 0\perp) \xrightarrow{\text{reduce}} \textcircled{4}$
 $\rightarrow (q_0, +, E\perp) \xrightarrow{\text{shift}} \textcircled{5}$
 $\rightarrow (q_0, \varepsilon, +E\perp)$ *we're now stuck!*

Can we parse $0+1 \times 2$

$(q_0, 0+1 \times 2, \perp) \xrightarrow{\text{shift}}$
 $\rightarrow (q_0, +1 \times 2, 0\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, +1 \times 2, E\perp) \xrightarrow{\text{shift}}$
 $\rightarrow (q_0, 1 \times 2, +E\perp) \xrightarrow{\text{shift}}$
 $\rightarrow (q_0, \times 2, 1+E\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, \times 2, E+E\perp) \xrightarrow{\text{shift}}$
 $\rightarrow (q_0, 2, \times E+E\perp) \xrightarrow{\text{shift}}$
 $\rightarrow (q_0, \varepsilon, 2 \times E+E\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, \varepsilon, E \times E+E\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, \varepsilon, E+E\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, \varepsilon, E\perp)$
 $\rightarrow (q_1, \varepsilon, \varepsilon)$

we could have done this

$\leadsto (q_0, \times 2, E+E\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, \times 2, E\perp) \xrightarrow{\text{shift}}$
 $\rightarrow (q_0, 2, \times E\perp) \xrightarrow{\text{shift}}$
 $\rightarrow (q_0, \varepsilon, 2 \times E\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, \varepsilon, E \times E\perp) \xrightarrow{\text{reduce}}$
 $\rightarrow (q_0, \varepsilon, E\perp)$
 $\rightarrow (q_1, \varepsilon, \varepsilon)$

we've parsed $0+[\times 2]$

we've parsed $[0 +] \times 2$

Because this grammar is ambiguous, we could shift to the left and reduce to the right and obtain two different results.